

Orange(Yingzhi) Ao

✉ ya2538@columbia.com

☎ +1 917-518-4578

🔗 Visit My GitHub

🌐 Visit My LinkedIn

EDUCATION

Nanjing University

B.A., Financial Engineering

2021-2025

Ranking: 3/31 | GPA: 92/100

Columbia University

Exchange Student

2024

GPA: 4.17/4.33

RELEVANT COURSES

CS And Stats

**DS in Finance And Insurance [A+], Statistical Machine Learning [A], Python, Applied Statistics II: Time Series, *Financial Engineering, *Financial Time Series (in R), *Artificial Intelligence, *Data Mining

Mathematics

**Stochastic Processes [A+], Numerical Analysis and Application Software (in MATLAB), Calculus, Discrete Mathematics [top 1], Linear Algebra, ODE, *PDE, *Financial Mathematics

* is added before courses in progress;

** is added before grad-level courses.

RESEARCH EXPERIENCES

Multi-Period Compliance MFG - FBSDEs in REC Market

RA

2024

Columbia University Stats Department

Supervised by Prof. Campbell, I set up NN models with Deep FBSDE Solvers for the multi-period MFG model, as an extended topic to the single-period model. For the next step, we will make it more realistic by adding more periods, inventory carryforward, and expiration rules based on the 2-period model.

- ✓ **Innovative Numeric Tricks** We proposed 3 tricks for greater stability and better convergence:
 - Clamping - restrict values within $[0, 1]$ and avoid gradient explosion;
 - Sigmoid Approximation - increase differentiability and continuity;
 - Logit Transformation - map $[0, 1]$ to $\mathbb{R}$ while providing differentiability.
- ✓ **Experimenting And Tuning** Experimented with various combos of optimizers, schedulers, and loss functions, tuning parameters like learning rate penalty weights for stable convergence.
- ✓ **Benchmarking** Proposed Separate Optimization as a benchmark, which could be interpreted as a short-sighted perspective against long-term planning by agents in the Joint Optimization.
- ✓ **Documentation, Typesetting, and GitHub-Publishing** Wrapped the models into well-structured packages and composed READMEs for code instructions. Summarized the period results into an [Overview Report](#). All works are documented and made public in [my GitHub repository](#).

Machine Learning In Insurance Pricing And Risk Modelin

RA

2024

Columbia University Actuarial Department

- ✓ **Team Collaboration** Collaborated with a team under the supervision and mentoring of Carlos (Arocha Association) to carry out an innovative model for a data-driving insurance pricing strategy.
- ✓ **Innovation** Originally applied K-Means to identify risk clusters using diverse policy data. Developed exposure rating models to estimate risk premiums and determine prices for each risk cluster.
- ✓ **Professional Reporting And Presenting** Delivered professional research plans, agendas, statements of work, and reports. Presented the findings and pricing strategy advice to clients.

Low-carbon Transition of China's Power System

RA

2023

Tsinghua-CTG Joint Center for Climate Governance and Low-carbon Transformation

Supervised by the lead researcher Yang Qu, I assisted her team in composing the research report series "Low-Carbon Transition of China's Power System". Key responsibilities and contributions are:

- ✓ **Co-Authorship And Editorial Support** Besides drafting, reviewing, and proofreading the final versions of the reports, I mainly researched and co-authored two major sections of the report series.
- ✓ **Data Collection And Visualization** I gathered and organized data and references for the report series. Also created all figures and tables for the whole series to provide data-driven insights.

PROJECT EXPERIENCES

Paired Trading Optimization with Genetic Algorithm 2023
Innovated a double-bound Bollinger strategy which significantly reduced fluctuations and turnover rate. Then applied Genetic Algorithms (GA) to optimize pair weights with customized fitness functions.

WorldQuant Brain Consultant | Alphathon (Gold) 2023
Secured a gold medal (11th place out of 200+ participants) in the Brain x NJU Alphathon 2023. Also contracted as a consultant to provide alpha ideas and received allowances.

INTERNSHIP EXPERIENCES

Higgs Assets High-Freq Quant Researcher 2024.08 - Present
Predicted correlations and developed a peer-grouping system, which will be applied to the multi-attention model in future steps. Working on tuning and packaging the grouping system at present.

TrexQuant Investment GAR Alpha Researcher 2024.06 - 2024.08
Developed alphas based on studies in behavioral finance. Applies regression and clustering techniques.

Chongwen Quantitative Quant Strategy Researcher 2023.09 - 2024.02
Adapted and optimized the Whitney George Strategy to the Chinese stock market, which has been used for trading since 2024 Q3. Simulated a trading system generating re-balancing instructions for traders.

SKILLS

Programming
Proficient in PyTorch and experienced in deep learning models. Versed in markdown, $\text{\LaTeX}$ with Overleaf, and Git commands with GitHub. Also know R, C++, and MATLAB basics through coursework.

Academic And Professional Typesetting
Experienced in delivering academic and professional reports, statements of work, and presentations.

English Proficiency
Scored 111/120 in TOEFL and 333/340 in GRE, with overseas experience at Columbia University.

Team Collaboration And Communication
Adaptable in a team through academic and student activities. Skilled in providing constructive feedback.